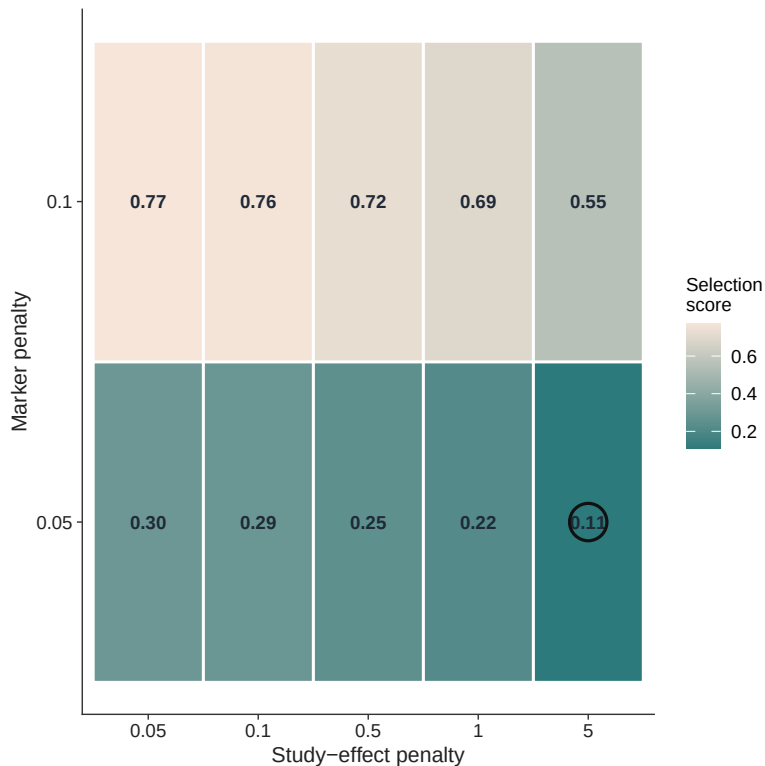


A

### Penalty-selection heatmap

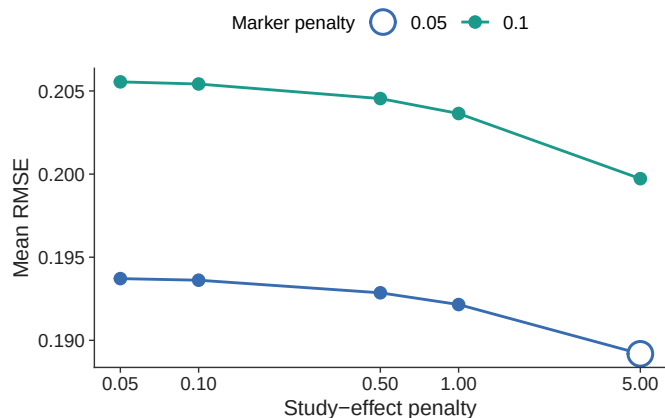
Lower score preferred; ring marks selected grid point



B

### State-recovery calibration

Stronger study shrinkage lowers RMSE



C

### Study-effect magnitude is regularized

